BLOCKCHAIN INVESTIGATION CASE STUDY

Public Trail Analysis of the Bybit 2025 Incident

An independent, open-source reconstruction of one delimited fund trail — and a measurement of where open-source analysis reaches its limit

CASE REFERENCE

BIC-2026-001

ANALYST

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INVESTIGATION PERIOD

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REPORT VERSION

1.0 — Final

CLASSIFICATION

Academic Portfolio

SUBJECT

Anchor 0x47666Fab…86E2, Ethereum mainnet

ANALYTIC STANDARDS

FATF 2020 Red Flag Indicators · ICD 203 · Admiralty Code (STANAG 2511) · ISO/IEC 27043

Contents

0. Revision History

This report is issued under version control. Once a version is released, subsequent changes are published as a new version with the modifications stated explicitly, rather than applied silently.

Version	Date	Summary
1.0	September 4, 2026	Initial release. Consolidates the findings of investigation phases 00–06, which were committed to version control progressively and are preserved in the project repository with their original timestamps.

1. Executive Summary

This report documents an independent reconstruction of a single fund trail associated with the publicly reported Bybit incident of February 2025, conducted using exclusively open-source tooling — a public block explorer and a purpose-built retrieval script — and no commercial blockchain intelligence platform.

The investigation was designed to answer a specific methodological question rather than to solve the case: **how far can an independent analyst trace such a trail with public data alone, and at what point does the method stop working?** The destination of the funds is already documented by parties with far greater resources; the distance an open-source analyst can cover, and the precise nature of the boundary that stops them, is not.

What was done

Five hops were traced from a publicly disclosed anchor address, under a stopping rule fixed in writing and committed to version control *before* any transaction data was examined. Every branch not followed was recorded with its address and value. Every claim produced was classified as fact, inference, or third-party attribution, and every inference carries an explicit confidence level.

Published analyses of the incident were **not consulted at any point** during the investigative phases. They were read only after all findings were finalized and committed, so that the comparison would measure independent reach rather than reading comprehension.

Principal findings

Finding	Detail
Trail reconstructed across five hops	From the anchor through derivative consolidation, conversion to native ETH via three decentralised exchanges, two levels of fragmentation, and a final consolidation point — approximately 98,376 ETH-equivalent at the first hop, narrowing to 454 ETH at the fifth.
Deliberate contamination of the transaction record identified	Nine impersonating token contracts using non-Latin characters that render identically to legitimate asset symbols, combined with 31 lookalike destination addresses matching genuine counterparties in their first six and last four characters. Together these form a specific trap capable of diverting a trace onto a fabricated path with no visible error.
Two FATF red-flag indicators identified, verified at source	Multiple high-value transactions in short succession (§11, p. 6) and conversion between virtual assets without logical business explanation (§12, p. 9). Four further observed patterns had *no* direct correspondence in the framework and are recorded as analytical observations rather than forced into a classification.
Independent trace corroborated by a specialist firm	The reconstruction matches SlowMist's published MistTrack analysis transfer-for-transfer through the first three hops — same addresses, same amounts, the stETH figure agreeing to four decimal places — despite that analysis having been read only after the findings were frozen.
The boundary of open-source analysis located precisely	Open-source tracing and a proprietary platform agree completely on the Ethereum-native path. They diverge at exactly one point: the cross-chain bridge. Crossing it requires proprietary heuristics or legal process — capabilities unavailable to open-source analysis by definition.
No attribution asserted	Published sources attribute the incident to a named actor on the basis of cross-incident clustering and off-chain intelligence. This analysis had neither and made no attribution — a restraint the subsequent comparison confirmed as correct.

> THE CENTRAL RESULT
>
> On the segment where the data is public and the path is linear, an independent analyst using a free block explorer reproduced a specialist firm's proprietary-tool findings **exactly**. The two part company at the cross-chain boundary, and only there. The limit encountered is not one of skill, effort, or tooling quality — it is the point at which the information required ceases to exist on the public Ethereum ledger.

Scope of what follows

The trail documented here carries approximately **0.11%** of the value that left the anchor address. The remaining 99.89% was not examined. No statement in this report about "the funds" should be read as applying to more than the specific path recorded in Section 7.

2. Objective and Scope

2.1 Objective

To independently reconstruct a single, delimited trail of funds associated with the publicly reported Bybit incident of February 2025, using exclusively open-source tooling, and to establish:

1. How far an independent analyst can trace such a trail without access to proprietary attribution data;
2. Which laundering typologies are observable in that trail;
3. Where the boundary between observable fact, analytical inference, and actor attribution falls in practice.

2.2 In scope

- One fund trail, from a publicly disclosed anchor address forward, under the stopping rule in Section 3.1.
- Transaction-level documentation of each hop: hash, timestamp, source, destination, asset, and amount.
- Identification of movement patterns and their correspondence, where any exists, to catalogued typologies.
- Reliability classification of every claim, and confidence levels for every inference.
- Comparison against published third-party analyses — conducted only after all findings were finalized.

2.3 Out of scope

- **Reconstruction of the full incident.** The sum involved was fragmented across a large number of addresses over an extended period. Tracing it in full is the work of teams with proprietary datasets and dedicated automation. No attempt at completeness was made and none is claimed.
- **Actor attribution.** No address is claimed to belong to any named individual, organization, or state.
- **Identification of natural persons.** No attempt was made to link any address to a real-world identity.
- **Discovery or publication of undocumented infrastructure.** Any address, pattern, or relationship surfaced that was not already present in public reporting would have been withheld and reviewed privately before any disclosure decision. No such material arose.
- **On-chain interaction of any kind.** This investigation is strictly read-only.
- **Inbound analysis.** Only outbound movement from each hop was examined. Two unexplained inbound discrepancies are recorded in Section 12 rather than pursued.

3. Methodology and Analytic Standards

3.1 Pre-registration of the stopping rule

The rule governing where the trace would begin, which branch would be followed at each fragmentation, and where it would end was written and committed to version control **before any transaction data was examined**.

> STOPPING RULE, AS PRE-REGISTERED
>
> **Branch selection.** Where funds fragment across multiple destinations, the branch carrying the highest value is followed. All other branches are recorded with their addresses and values.
>
> **Tie-breaking.** Where two or more branches carry identical amounts, the branch whose transfer bears the earliest timestamp is followed; where timestamps are also identical, the lowest destination address in lexicographic order.
>
> **Termination.** Tracing stops at whichever occurs first: the traced funds enter a mixing service or cross-chain bridge, or five hops have been traced from the anchor.

Why this was fixed in advance. Without a rule declared beforehand, an investigator stops when they tire and afterwards constructs a justification for the endpoint they happened to reach. The two outcomes produce identical text in a report and are indistinguishable to a reader. Pre-registration, with the commit history to evidence it, converts the endpoint from a point of exhaustion into a methodological decision that a third party can audit.

The rule was consequential in practice. At Hop 3 the trace encountered nine branches of exactly 10,000 ETH each — a condition under which "highest value" selects nothing. The tie-breaking criterion, written before that condition was encountered, resolved the choice without discretion being exercised at the point of decision.

A modification, recorded rather than applied silently

The tie-breaking criterion was *added* to the original rule after anchor validation revealed forty branches of identical value, and before any hop was traced. Both the addition and its reason are recorded in the project's revision log. It is disclosed here because a rule amended in silence is indistinguishable from a rule invented after the fact.

Scope extension: available, unused

The pre-registered rule permits extension to additional branches under one condition only — where the followed branch terminates prematurely, for a reason unrelated to the stopping rule. Extension is not available because a result was uninteresting. Had it been exercised, all branches traced would have been reported, including any producing less informative results. **It was not exercised.**

3.2 Claim taxonomy

A blockchain ledger produces an unusual evidentiary situation: the raw data is more reliable than in almost any other forensic domain — immutable, timestamped, verifiable by any party against any node — while conclusions drawn from it are frequently weaker than they appear, because the link between an address and a controlling entity is almost never directly observable.

This creates a specific failure mode: an analyst presents a well-supported fact and an unsupported attribution in the same sentence, in the same tone, and the reader cannot tell them apart. Every claim in this report is therefore assigned one of three types, stated explicitly wherever the claim appears.

Type	Definition	Evidentiary basis
FACT	A record that exists on the ledger and can be verified independently by any third party	The blockchain itself — immutable and falsifiable by anyone
INFERENCE	An analytical conclusion drawn from observed facts by applying a stated heuristic	Reasoning over facts. May be well supported and still wrong. The heuristic is stated so it can be challenged
ATTRIBUTION	A claim linking an address to a named real-world actor	Requires evidence outside the ledger — exchange records, off-chain intelligence, legal process. **Not asserted by this analysis**

Relationship to the classical forensic classification

Conventional digital forensics classifies evidence as *primary*, *corroborative*, or *circumstantial*. That taxonomy grades the **strength** of an item of evidence relative to a proposition. The taxonomy applied here grades the **epistemic type** of a claim — whether it is read from the record, reasoned from it, or imported from outside it.

The distinction matters in this domain specifically. On a public ledger, the primary record is universally available and beyond dispute; nothing is gained by grading a transaction hash as "primary evidence," because all transaction hashes are equally primary. What varies, and what a reader needs to know, is whether a given statement *is* a transaction record, an interpretation of several, or an assertion by a third party. The two schemes correspond approximately as follows:

Classical	Applied here	Correspondence
Primary evidence	FACT	A direct, first-party record of the act itself — on a public ledger, the transaction record
Corroborative evidence	INFERENCE — HIGH	Independently supports a conclusion through convergence of multiple observations, without being the act itself
Circumstantial evidence	INFERENCE — LOW	Consistent with a conclusion, but a competing explanation is roughly as viable
(no direct equivalent)	ATTRIBUTION	Isolated deliberately: in this domain the most common error is presenting a third-party label as though it were a record

3.3 Confidence levels

Applied to inferences only, adapted from ICD 203 analytic tradecraft standards.

Level	Meaning
HIGH	The conclusion follows directly from observed data. Alternative explanations exist in principle but are not credible given the evidence.
MODERATE	Well supported, but dependent on a heuristic that can fail. Alternatives exist and are less likely, but not implausible.
LOW	Consistent with the observed data, but the evidential basis is thin or a competing explanation is approximately as viable. Recorded for completeness; not relied upon in any conclusion.

3.4 Standards applied, and where

Standard	Application in this investigation
FATF (2020) *Virtual Assets: Red Flag Indicators of Money Laundering and Terrorist Financing*	Typology classification (Section 9). Consulted directly; where a pattern corresponds to an indicator, the section and page are cited so the correspondence can be verified in the source rather than trusted from this report's paraphrase.
ICD 203 *Analytic Standards*	Separation of observed data from analytic judgment; expressed confidence levels (Sections 3.2, 3.3, 9).
Admiralty Code NATO STANAG 2511	Grading of every external source consulted, for reliability (A–F) and credibility (1–6). Applied to the anchor provenance (Section 5) and to the published analyses (Section 11).
ISO/IEC 27043 *Investigation principles and processes*	The preparation stage: scope, objectives and constraints defined before collection began (Section 2), which is what separates an investigation from an undirected sweep and provides the basis for deciding when collection is complete.
Lei 12.737/2012, Art. 154-A Brazilian Penal Code	The passive-collection posture (Section 13) was chosen so that no query touches a target system — the statutory boundary between consulting published records and unauthorized access.
LGPD / GDPR	Considered and found to have limited application: blockchain addresses are not, without further linkage, personal data, and no attempt at such linkage was made. The relevant principle applied is purpose limitation — no enrichment or identification was attempted (Section 13).

3.5 Sequence integrity

Published analyses of this incident exist and are readily available. They were **not consulted during the investigative phases**. Only the anchor address was sourced externally, and the project's collection log records precisely what was read, when, and from where.

Cross-validation occurred only after all findings were finalized and committed. The reasoning is that an analyst who reads the answer first does not investigate — they search for confirmation, and cannot afterwards distinguish what they found from what they were looking for. Preserving the separation is what makes the comparison in Section 11 meaningful: it measures independent reach, not reading comprehension.

The findings were not revised after the comparison. Where a published source bears on a finding, that is recorded in Section 11 as an outcome of comparison, and the frozen finding stands as written.

4. Operational Security Considerations

Most laboratory forensics carries no risk to the examiner. This investigation differs in one respect: the incident under study has been publicly attributed by the FBI to a state-linked threat actor with a documented history of targeting security researchers.

The recorded pattern is not retaliation against investigators. It is opportunistic compromise of a demographic — false recruiter profiles on professional networks, fabricated collaboration offers, malicious repositories presented as tooling, and weaponized datasets offered for analysis. The target profile is a security professional who is visibly active and open to opportunity. Publishing work on this subject increases visibility within that demographic.

The measures below are therefore proportionate to a realistic threat model, not a dramatic one. This investigation reads a public ledger and executes nothing.

4.1 Threat model

Vector	Assessed likelihood	Rationale
Targeted action against the analyst in response to this analysis	**Very low**	The trail examined is already documented publicly by specialist firms. This work reveals nothing new and presents no operational threat to any party.
Opportunistic social engineering following publication	**Plausible**	Consistent with documented campaigns against security researchers. Not specific to this project — the analyst already fits the target profile — but publication increases visibility.
Passive correlation of query activity	**Low**	Queries are directed at public explorer services, not at infrastructure under adversary control. Mitigated by the egress control below.
Compromise via on-chain interaction	**Eliminated by design**	No wallet connected and no transaction broadcast at any point. There is no interactive surface, which also eliminates address-poisoning and dusting vectors against the analyst.

4.2 Controls adopted

Control	Implementation	Purpose
Isolated analysis environment	Dedicated virtual machine, snapshotted clean before first use	Separates research activity from the primary working system
Network egress control	VPN active for all explorer queries and API calls	Prevents association of query activity with the analyst's residential address
Disposable credentials	Burner email created within the isolated environment, used for all tooling accounts	Prevents linkage between tooling accounts and the analyst's primary identity
Read-only posture	No wallet connected; no contract interaction; no transaction broadcast	Eliminates the interactive attack surface entirely
No execution of third-party code	Repositories, scripts, datasets and documents from unverified sources are not executed, opened, or imported — in any environment	Addresses the highest-likelihood vector at zero technical cost
Unsolicited approaches treated as untrusted	Following publication, contact offering opportunity, collaboration, datasets or tooling is treated as hostile until independently verified through a channel not controlled by the approaching party	The control that matters most; it does not expire, because publication increases exposure indefinitely

4.3 Verification, and a silent failure

The VPN client's `Connected` status was **not** accepted as evidence that traffic was tunnelled. Verification was performed by direct inspection: the observed exit address was confirmed on both IPv4 and IPv6 protocol stacks; the tunnel interface was confirmed to exist and be up; and the policy-routing rules directing non-marked traffic through the tunnel were confirmed present.

> **WHY THIS MATTERED**
>
> An earlier connection attempt reported `Connected` at the client level and displayed a foreign exit IP address, while the routing table still showed the default route via the local network adapter and the resolver was still the residential ISP's. No tunnel interface had been created. Root cause: a second, host-only network adapter retained from earlier laboratory work caused the tunnel backend to fail interface assignment. **The failure was silent at the client level and visible only through direct inspection.** Had the client's status message been trusted, every query in this investigation would have egressed unprotected while the record claimed otherwise.

4.4 Controls considered and rejected

Documenting what was rejected, and why, is part of an honest threat model. Security theatre is not security.

Considered	Adopted	Reasoning
Routing all traffic over Tor	No	VPN egress control meets the actual requirement. Many explorer services degrade or block Tor exit nodes, and the marginal benefit does not justify the operational cost for read-only public-ledger queries.
Non-persistent / amnesiac VM	No	Appropriate where untrusted code is executed. This investigation executes nothing.
Dedicated physical hardware	No	Disproportionate to read-only analysis of public data.
Anonymous publication	No	Would defeat the professional purpose of the work. Attribution of authorship is an accepted trade-off — which is precisely why the behavioural controls above are treated as permanent.

An analyst who investigates a sophisticated threat actor without modelling their own exposure has completed only half the analysis.

5. Anchor Point and Provenance

The trace requires a starting address. Obtaining it necessarily involves reading external sources — the discipline lies in stopping at the right point. Identifying which address was publicly disclosed as involved is permitted; reading any analysis of where the funds subsequently went is not, since that is the question this investigation exists to answer independently.

5.1 The anchor

Address	`0x47666Fab8bd0Ac7003bce3f5C3585383F09486E2`
Chain	Ethereum mainnet (chain ID 1)
Explorer label observed	"Bybit Exploiter" — *third-party attribution, see 5.3*
Date sourced	2026-08-24
Basis for selection	Outbound activity concentrated on 2025-02-21, in assets and at a scale consistent with the publicly reported incident (validated at 5.4)

5.2 Provenance chain

The anchor was reached through three steps of decreasing dependence on any external claim. They are recorded separately because they carry different evidentiary weight.

Step	Source	Grading	Treatment
1	AI-generated search summary, which cited a block explorer as its source	**F6**	Not a citable source; cannot be audited; reliability cannot be assessed. Treated as a **navigation shortcut to a source it named**, never as evidence.
2	Etherscan — address record	**B2**	The address was confirmed directly at the named source rather than accepted from the summary. Rated B as an intermediary presenting protocol data; its *labels* are its own editorial attributions.
3	On-chain behaviour of the address itself	—	The step the anchor actually rests on. Not a source in the intelligence sense; the primary record.

WHAT THE ANCHOR RESTS ON

The anchor rests on step 3. Steps 1 and 2 determined where to look; step 3 determined whether it held. Had the pointer been wrong, validation would have rejected it regardless of what any source claimed. The weak source is recorded rather than omitted: a weak source that is disclosed and neutralised is manageable, while one concealed behind a stronger-sounding citation is not.

5.3 The explorer label is not a fact

The explorer displays this address as "Bybit Exploiter". Nothing in the protocol identifies an address as belonging to any actor; the label is an editorial assertion by the explorer. It is recorded because it informed selection, and is treated throughout this report as an attribution claim — never as a finding of this analysis. Explorer labels are also perishable, and captures were taken for that reason.

5.4 Behavioural validation

The address was queried directly and its transaction history examined for consistency with the reported incident.

Asset	Contract	Outbound total
ETH (native)	*protocol asset*	400,001
stETH	`0xae7ab96520de3a18e5e111b5eaab095312d7fe84`	90,375.55
mETH	`0xd5f7838f5c461feff7fe49ea5ebaf7728bb0adfa`	8,000
cmETH	`0xe6829d9a7ee3040e1276fa75293bde931859e8fa`	15,000

Anchor validation

OBSERVED Outbound movement of approximately 400,000 ETH concentrated on 2025-02-21, alongside three liquid-staking derivatives. FACT

INTERPRETATION The date, scale and asset composition are consistent with the publicly reported incident. HIGH

BOUNDARY Consistency with a reported incident is not proof of involvement in it. This establishes the address as a plausible anchor for the trail. It does not establish who controlled it, and no such claim is made.

5.5 A note on address handling

The anchor address was initially mis-transcribed on copy, producing a 41-character string. The retrieval tool rejected it on format validation before any query was issued. This is recorded because it is instructive: a 42-character hexadecimal address carries no human-readable redundancy, and a single dropped character produces a string that looks entirely plausible. Address handling in this investigation is by copy-and-validate, never by transcription, and format validation in tooling is a cheap control against a failure mode that would otherwise be silent.

6. Consolidated Timeline

All times UTC. The **Source** column distinguishes what this analysis established from what is reported by third parties; the **Interpretation** column states what each event establishes for the investigation.

Timestamp	Event	Source	Interpretation
2025-02-19 07:15:23	Malicious implementation contract deployed	*Third party (SlowMist)*	Preparation preceding the incident — not established by this analysis
2025-02-21 14:13:35	Safe contract logic replaced via three owner signatures	*Third party (SlowMist)*	The incident itself — not established by this analysis
2025-02-21 14:29:47	Anchor sends 1 ETH to `0xa4b2fd68…449e`	This analysis	Precedes the asset transfers; magnitude consistent with fee funding at the destination
2025-02-21 14:41:23	10,000 stETH to `0xa4b2fd68…449e`	This analysis	Derivative consolidation begins
2025-02-21 14:43:23	First stETH tranche to a DEX aggregator contract	This analysis	Conversion begins within two minutes of receipt
2025-02-21 14:45:11	9,980 ETH to `0xdd90071d…5f92`	This analysis	First conversion proceeds forwarded onward
2025-02-21 14:47:11	50,000 stETH to `0xa4b2fd68…449e`	This analysis	Largest single derivative transfer of the trace
2025-02-21 14:56:11	First of forty 10,000 ETH transfers from the anchor	This analysis	Native-ETH fragmentation begins — a branch not followed
2025-02-21 15:10:35	8,000 mETH to `0xa4b2fd68…449e`	This analysis	Second derivative type consolidated to the same destination
2025-02-21 15:12:23	30,375.55 stETH to `0xa4b2fd68…449e`	This analysis	Derivative consolidation completes
2025-02-21 15:23:35	15,000 cmETH to `0x1542368a…4443`	This analysis	Branch not followed; subsequently reported frozen and recovered by the issuer (§10)
2025-02-21 15:27:47	Final stETH tranche to the DEX aggregator (eighth of eight)	This analysis	Conversion completes across 44 minutes
2025-02-21 15:32:35	38,313.79 ETH to `0xdd90071d…5f92`	This analysis	Conversion proceeds fully forwarded; inflow and outflow reconcile to 0.333%
2025-02-21 15:48–15:54	Bulk of the forty-way native fragmentation	This analysis	400,000 ETH divided in a six-minute window
2025-02-21 16:04:23	First of nine 10,000 ETH transfers → `0xf3025725…72be`	This analysis	Second fragmentation level; the followed branch, selected by the pre-registered tie-breaker
2025-02-21 16:05:11	Ninth and final 10,000 ETH transfer	This analysis	90,000 ETH divided in 48 seconds
	— nine days of dormancy —		
2025-03-02 14:45:59 – 17:01:11	74 outbound transfers to 58 destinations, irregular amounts	This analysis	Third fragmentation level, with a changed mechanism: dispersion rather than division into blocks
2025-03-03 15:42:59 – 18:13:47	45 further transfers from the Hop 3 address	This analysis	The residual 10% of that address's holdings, moved a day after the parallel dormancy ended
2025-02-27	FBI attributes the incident to a named actor	*Third party (FBI via AP)*	Attribution by a party with capabilities this analysis does not have — not a finding of this analysis

The two dormancy periods are near-simultaneous: the Hop 3 address moved on 2025-03-03 and the Hop 4 address on 2025-03-02, each approximately nine to ten days after receipt. This is addressed as a pattern in Section 9.

7. Trace Findings

Five hops were traced. Each entry records what was retrieved, how signal was separated from contamination, which branch was selected and on what basis, and — critically — which branches were *not* followed. A branch that is documented is a scope decision; a branch that is omitted is a gap.

7.0 What counts as a hop

Not every destination is a counterparty. Funds sent to a smart contract that performs a swap have not arrived anywhere — they have been converted, and the trail continues in whatever asset emerged. Treating such a contract as a destination would end the trace at infrastructure rather than at a party.

Every destination in this trace was therefore checked on the block explorer to establish whether it is a contract or an externally-owned account before being treated as a hop. Where a destination proved to be a swap venue, the continuation was identified by reconciling what entered against what left.

7.1 Valuation basis, declared before tracing

The anchor's outbound movement spans four legitimate assets. The pre-registered rule selects the highest-value branch, which requires comparing amounts across assets. **stETH, mETH and cmETH are treated as approximately 1:1 equivalent to ETH** for the purpose of branch selection; all three are ETH liquid-staking derivatives designed to track the value of the underlying asset.

This is a stated judgment, not a fact. These instruments trade at a small and variable discount or premium to ETH, and no market data was consulted to establish the rate at the relevant time. It is recorded because the branch selected at Hop 1 depends on it and a reader is entitled to disagree. It is sufficient for ranking branches whose magnitudes differ by roughly an order of magnitude; it would not be sufficient where two branches were close in value.

7.2 Hop 1 — Derivative consolidation

From `0x47666Fab8bd0Ac7003bce3f5C3585383F09486E2` (anchor) · **Retrieved** 2026-08-26 14:10:09 UTC · 255 transfers (46 native, 209 token) · no filter applied at collection.

Of twenty distinct token groups returned, only three are legitimate assets. The remaining seventeen are impersonating contracts (Section 8). All were excluded from analysis but retained in the evidence data.

Destination	ETH-equivalent	Composition
`0xa4b2fd68593b6f34e51cb9edb66e71c1b4ab449e`	**98,376.55**	1 ETH · 90,375.55 stETH · 8,000 mETH
`0x1542368a03ad1f03d96d51b414f4738961cf4443`	15,000.00	15,000 cmETH
40 distinct addresses	10,000.00 *each*	ETH (native)

Two handling patterns in the same hour

OBSERVED Between 14:29:47 and 15:54:23 the anchor made 46 outbound transfers of legitimate assets. One destination received 98,376.55 ETH-equivalent across five transfers in four assets; a second received 15,000 cmETH; the remaining 400,000 ETH was divided into exactly forty transfers of 10,000 ETH each, most within a six-minute window. FACT

INTERPRETATION Consolidation of the staked-ETH derivatives into a single destination, and uniform fragmentation of the native ETH across forty — two distinct handling patterns applied concurrently. HIGH as to the pattern.

BOUNDARY Establishes what moved, when, and to which addresses. Does not establish who controlled any address, whether the forty destinations are under common control, or why the two asset classes were handled differently. The derivatives may have been consolidated for reasons intrinsic to those instruments — redemption requires interacting with their protocols — rather than for any concealment purpose.

Branch selected: `0xa4b2fd68…449e`, exceeding the next largest by a factor of approximately 6.5 and each native-ETH branch by approximately 9.8. The margin is wide enough that the outcome does not turn on the exchange rate assumed at 7.1.

Branches not followed: 41, carrying approximately 415,000 ETH-equivalent — the forty native-ETH destinations and the cmETH destination. All are recorded with addresses and transaction hashes in the evidence data. **The followed branch carries roughly 19% of what left the anchor.**

7.3 Hop 2 — Conversion via decentralised exchanges

From `0xa4b2fd68593b6f34e51cb9edb66e71c1b4ab449e` · **Retrieved** 2026-08-26 14:42:21 UTC · 43 transfers (20 native, 23 token).

Destination	Received	n	Type
`0xdd90071d52f20e85c89802e5dc1ec0a7b6475f92`	98,048.7948 ETH	3	Account
`0x6bb00006700545070400310063 2eb93ea00c0000`	75,187.7740 stETH	8	**Contract** — DEX aggregator
`0xfe837a3530dd566401d35befcd55582af7c4dffc`	15,187.7740 stETH	1	**Contract** — DEX router
`0x04708077eca6bb527a5bbbd6358ffb043a9c1c14`	8,000.0000 mETH	3	**Contract** — liquidity pool
`0x1542368a03ad1f03d96d51b414f4738961cf4443`	0.1 ETH	1	—

Each destination receiving a derivative was verified on the block explorer. All three are contracts, and all three are decentralised-exchange infrastructure — one tagged as a liquidity pool, one as a fee-route proxy with transaction methods listed as swap operations, and one whose creator carries an aggregator protocol tag. These labels are third-party attributions and are perishable; captures were taken.

Conversion established by reconciliation

OBSERVED Received 98,376.55 ETH-equivalent in derivatives. Sent 90,375.55 stETH and 8,000 mETH to three exchange contracts, and 98,048.79 native ETH to a single account — all within approximately 50 minutes. FACT

RECONCILIATION Inflow 98,376.5479 · native outflow 98,048.8948 · **difference 327.6531, or 0.333%**

INTERPRETATION The derivatives were converted to native ETH through decentralised exchanges and the proceeds forwarded onward. HIGH — two independent lines converge. **Arithmetically:** a 0.333% margin is consistent with exchange fees and slippage; had the derivatives been transferred to third-party wallets *in addition to* the ETH leaving, total outflow would have approximated twice the inflow, and it does not. **Structurally:** all three derivative destinations are exchange contracts rather than accounts, and assets sent to them are exchanged, not held.

BOUNDARY The reconciliation establishes that the amounts are consistent with conversion. It does **not** prove that the specific ETH forwarded onward is the same ETH that emerged from these particular swaps — fungibility makes that claim unavailable on-chain. What is established is that this address received derivatives, exchanged derivatives, and forwarded an almost identical quantity of ETH within the same window.

Tranching of the conversion

OBSERVED The stETH sent to the aggregator was divided into eight transfers over 44 minutes: 10,000 · 25,000 · 12,000 · 7,000 · 6,000 · 7,187.77 · 4,000 · 4,000. FACT

INTERPRETATION Dividing a large conversion into portions is standard practice for limiting price impact on a decentralised exchange, and is equally consistent with avoiding a single conspicuous transaction. LOW as to purpose — both explanations fit and the evidence does not distinguish.

Termination check. Decentralised exchanges are *not* a termination condition. The rule terminates at a mixer or cross-chain bridge, both of which sever the link between input and output. A swap does not: the output is observable on the same chain and reconciles against the input, as demonstrated. The trail continues.

7.4 Hop 3 — Two operating regimes, and the tie-breaker

From `0xdd90071d52f20e85c89802e5dc1ec0a7b6475f92` · **Retrieved** 2026-08-26 20:43:27 UTC · 284 transfers (54 native, 230 token).

No legitimate token transfers were present. All nine token groups are impersonating contracts — seven with non-Latin symbols and **two with the clean-ASCII symbol** `ETH` (contracts `0x2f22686d…d236` and `0xfabda805…7215`). There is no legitimate ERC-20 token with the symbol ETH on Ethereum mainnet; ETH is the protocol's native asset, not a token contract. Any ERC-20 presenting itself as ETH is by definition impersonating. Both were excluded on the analyst's judgment, not on any automated warning — the division of labour the tooling documentation states explicitly.

	Regime A	Regime B
Date	2025-02-21	2025-03-03
Window	16:04:23 – 16:05:11 (**48 seconds**)	15:42:59 – 18:13:47 (~2h 31m)
Transfers	9	45
Distinct destinations	9	30
Amounts	10,000 ETH exactly, all nine	Irregular: 0.88 to 561.06
Total	**90,000.0000 ETH**	8,055.3621 ETH

Two operationally distinct regimes

OBSERVED 90% of the value left in nine identical round-number transfers within 48 seconds. The remainder left ten days later, in 45 irregular transfers over two and a half hours, several destinations receiving multiple times. FACT

INTERPRETATION Two operationally distinct handling patterns applied to the same holdings: rapid bulk distribution in uniform units, then, after a pause, dispersed distribution in varied amounts. HIGH as to the distinction.

BOUNDARY The ten-day gap and the change of pattern are facts. Whether they reflect a single actor changing method, different actors, or an automated process with a scheduled second stage is **not established**.

Reconciliation discrepancy

OBSERVED Received 98,048.7948 ETH; sent 98,055.3621 ETH. Outflow exceeds traced inflow by **6.5673 ETH**. FACT

INTERPRETATION The address received ETH from at least one source other than the branch under trace. HIGH — the arithmetic admits no alternative.

BOUNDARY The additional inflow was not traced; inbound analysis is outside scope. Its magnitude (0.007% of throughput) is consistent with fee funding, but that is inference. It is recorded because an unexplained discrepancy that goes unmentioned is indistinguishable from one that went unnoticed.

The tie-breaker applied

Regime A produced nine branches carrying exactly 10,000 ETH each. The highest-value criterion does not discriminate between them. The pre-registered tie-breaking rule applied for the first time in this trace: the earliest timestamp. The transfer at **16:04:23** is unique — the next three occur twelve seconds later — so the secondary criterion (lexicographic ordering) was not required.

Branch selected: `0xf302572594a68aa8f951fae64ed3ae7da41c72be`, transaction `0x376e337568836889c535ef3947ea89bb918a091e3e6199eefdbe6a6fbacf4579`.

The rule was written before any hop was traced and before this condition was encountered. It resolved the selection without discretion being exercised at the point of choice, which is the entire purpose of pre-registering it.

Destination verification

Type	**Externally-owned account**, not a contract
Explorer label	"Bybit Exploiter 45"
Tags	`Exploit`, `# Bybit Exploit`
Warning banner	Reports that the address was used in an exploit, citing a named third-party investigator
Funded by	"Bybit Exploiter 5"
Balance at retrieval	0.00047 ETH — the funds moved on

ON THE SIGNIFICANCE OF THIS LABEL

This trail was followed from the anchor by mechanical application of the pre-registered rules — highest value, highest value, earliest timestamp — with no third-party analysis consulted at any point. The address arrived at by that route carries an independent label associating it with the same incident, and is recorded as funded by another address in the same labelled cluster.

This is corroboration of the path, not attribution. The label is a third-party assertion citing a third-party investigator; nothing in the protocol identifies an address as belonging to any actor. Nor does it constitute the Section 11 cross-validation, which compares against published analyses of fund movement — those remained unread. What is recorded here is an entity label encountered incidentally while verifying whether a destination was a contract or an account.

Branches not followed: eight further 10,000 ETH branches (80,000 ETH), tied with the selected branch and separated only by the tie-breaker, plus 45 Regime B transfers (8,055.36 ETH) across 30 destinations. **The followed branch carries approximately 10.2% of what left this address.**

7.5 Hop 4 — Dispersion, and convergence between branches

From `0xf302572594a68aa8f951fae64ed3ae7da41c72be` · **Retrieved** 2026-08-26 21:09:16 UTC · 147 transfers (74 native, 73 token).

The 10,000 ETH received on 2025-02-21 remained at rest for nine days, then left on 2025-03-02 in 74 transfers of irregular amounts (11.96 – 303.31 ETH) to 58 distinct addresses over roughly two hours. Total outflow 10,062.3582 ETH — exceeding the traced inflow by 62.3582 ETH, again indicating an untraced additional source. **No destination receives more than 4.5% of the total:** this is genuine dispersion, not division into blocks.

Convergence between separately-fragmented branches

OBSERVED Four addresses receive from **both** Hop 3 and Hop 4 — which sit on different levels of the same tree, Hop 4 being a child of one of Hop 3's nine tied branches. One of the four (`0x21032176…044c`) received one of the nine 10,000 ETH transfers at Hop 3 — a tied branch, not followed — and later receives again via the branch that *was* followed. FACT

INTERPRETATION The fragmentation observed at earlier hops is not divergent; the paths reconverge, which suggests the division was of routing rather than of destination. Repeated reconvergence is consistent with common control of the addresses involved. LOW

BOUNDARY Convergence does **not** establish common control. Two unrelated parties depositing at the same downstream service produce an identical signature. Distinguishing the two would require information about the receiving addresses that on-chain data does not carry.

Verification of the convergent addresses

Address	Label	Funded by	Onward behaviour
`0x8ed8553d…6036`	*none*	"Bybit Exploiter 9"	Transfers to a stablecoin protocol
`0x8b62111b…1ca1`	*none*	`0xF9Fe2410…59127`	**Deposit calls to THORChain**
`0x21032176…044c`	**"Bybit Exploiter 46"**, tagged `Exploit`	"Bybit Exploiter 5"	Transfers to "Bybit Exploiter 47"
`0x54acab84…fba2`	*none*	"Bybit Exploiter 42"	**Repeated deposit calls to THORChain**, receiving from addresses labelled "Bybit Exploiter 5", "52" and "53"

Two of the four deposit into THORChain, a cross-chain swap protocol — one systematically, in amounts of 208 to 223 ETH. A cross-chain bridge is the *primary* termination condition of this investigation's stopping rule, precisely because it severs the on-chain link between input and output: funds enter on Ethereum and leave on a different chain with no on-chain record joining the two sides.

The branch selected at this hop does not itself deposit into THORChain. But adjacent branches — reached from the same parent, carrying comparable amounts — do. The observation is recorded because it bears on where this trail was heading regardless of which branch the rule selected.

A disclosed interpretive choice

> **TWO READINGS OF THE RULE, AND WHY THE LESS CONVENIENT ONE WAS APPLIED**
>
> The pre-registered criterion selects "the branch carrying the highest value". At this hop the two readings diverge:
>
> **Branch as destination, aggregated:** `0x327ffe25…ba23`, 454.06 ETH across two transfers.
> **Branch as individual transfer:** `0x54acab84…fba2`, 303.31 ETH in one — which deposits into THORChain.
>
> The **aggregated reading was applied**, on the basis that a branch is a path the funds took, and two transfers to one address are one path. This is recorded because the alternative was available and would have produced a different — and, for this report, a distinctly more convenient — outcome: it would have terminated the trace at a bridge, a cleaner ending than the one actually reached.
>
> **Selecting the reading that produces the better result, after seeing where each leads, is precisely what pre-registration exists to prevent.** The interpretation applied follows from the rule's purpose, not from its consequences here.

Branch selected: `0x327ffe25f330d2d809106ace581099ad1af9ba23` — an unlabelled externally-owned account, recorded as funded by "Bybit Exploiter 19" and receiving from "Bybit Exploiter 45" and "46". **Branches not followed:** 57 destinations carrying 9,608.30 ETH — 95.5% of what left this address.

7.6 Hop 5 — Consolidation and termination

From `0x327ffe25f330d2d809106ace581099ad1af9ba23` · **Retrieved** 2026-08-28 16:50:17 UTC · 65 transfers (9 native, 56 token).

Six token groups, **all impersonating** — every one carrying a non-Latin symbol, and two contracts appearing for the first time at this hop, indicating the impersonation campaign extends across the address set rather than targeting individual addresses.

Destination	Received (ETH)	Transfers
`0xf2fa83097ebb54b5204300fd88df7cb1a803276a`	454.0717	2
`0xe75e053856692c390700cb1f09928e1a1c9e2019`	452.5749	2
`0x535c86ef69cd7f7c618409928aff610b7cab7b83`	444.9700	2
`0x5679b4b6ac7968d56972822e4ba9e0e9aa731021`	238.6969	1
`0x9d977674985e1aecd1b5162456e2429ccbce2bbd`	61.3400	1
`0x1743b4e13273ab8e86de51edaef12c5ef2026416`	0.3000	1

Traceability dissolution

OBSERVED This address sent **1,651.9535 ETH** while the traced branch delivered **454.0611 ETH** to it — 3.6 times more left than arrived by the path under trace. Its record shows inbound transfers from addresses labelled "Bybit Exploiter 45", "46" and "19". FACT

INTERPRETATION This address is a consolidation point, pooling funds arriving by several separate paths. HIGH

CONSEQUENCE ETH is fungible. Once funds from multiple paths are pooled in a single account, **no on-chain evidence distinguishes which outbound transfer carries which inbound funds**. Continuing would mean following a destination carrying, at most, a 27% probability of containing the funds under trace — and stating otherwise would be an assertion the evidence cannot support. **This is a limit of the evidence, not of the method:** no amount of additional tracing, tooling, or patience resolves it, because the information required does not exist on-chain.

Termination

The trace terminates under two independent conditions:

1. **Hop limit reached** — five hops from the anchor, the secondary termination criterion of the pre-registered rule.
2. **Traceability dissolved** — consolidation renders the traced funds indistinguishable from others at this address.

The *primary* termination condition — mixer or bridge entry — was not reached on the followed branch. It was, however, observed on adjacent branches at Hop 4. The trail this analysis followed did not itself reach that exit, but paths one step to either side of it did.

7.7 Trace summary and the proportion traced

Hop	Address	Received (traced)	Characteristic
Anchor	`0x47666Fab…86E2`	—	400,001 ETH plus derivatives; two concurrent handling patterns
1	`0xa4b2fd68…449e`	98,376.55 ETH-eq	Derivative consolidation
2	`0xdd90071d…5f92`	98,048.79 ETH	Post-conversion; received via DEX swaps
3	`0xf3025725…72be`	10,000 ETH	Labelled "Bybit Exploiter 45"; nine-way uniform split
4	`0x327ffe25…ba23`	454.06 ETH	Dispersion across 58 destinations
5	*terminated*		Consolidation point; traceability dissolved

THE SINGLE MOST IMPORTANT BOUNDARY IN THIS REPORT

The followed path carries a progressively smaller share of the funds leaving each address: approximately 19% at Hop 1, 10% at Hop 3, 4.5% at Hop 4. Of the roughly 415,000 ETH-equivalent that left the anchor, this trace follows a path ending in 454 ETH — **around 0.11%**.

This is the stopping rule operating exactly as designed. It is also the boundary on everything this report concludes: **no statement about "the funds" applies to more than the specific path documented above.** The remaining 99.89% was not analysed, and nothing here should be read as describing it.

8. Adversarial Contamination of the Transaction Record

The transaction record of every address on this trail is contaminated. The contamination is not incidental spam: it is constructed, at scale, in a form that specifically targets the method of analysis applied here.

8.1 Token symbol impersonation

A token's symbol is chosen by whoever deploys the contract. It is not an identifier, and carries no guarantee of uniqueness or honesty. Retrieval at the anchor returned twenty distinct token groups; only three were legitimate assets.

Homoglyph impersonation

Nine groups carry symbols containing characters from non-Latin alphabets that render identically to Latin letters — Cyrillic `Е` (U+0415) for `E`, `Т` (U+0422) for `T`, `ѕ` (U+0455) for `s`. In a rendered table these are visually indistinguishable from the assets they imitate:

Symbol as displayed	Transfers	Nature
ETH	57	Cyrillic E, T, H — not ETH
ETH *(native)*	52	**Legitimate** — the protocol asset
ETH	45	Cyrillic E
ETH…	35	Cyrillic E, T, H
ETH	27	Cyrillic T
stETH	17	Cyrillic T
stETH	11	**Legitimate** — Lido contract
ѕтETH	2	Cyrillic ѕ, т, H

One further token carries the clean-ASCII symbol `BybitExploiter`, with 17,330,506 units transferred **to the queried address itself** — consistent with deliberate contamination of a notable address's visible history.

Clean-ASCII impersonation, which no automated check catches

At Hop 3, two contracts presented the symbol `ETH` in unaltered ASCII. There is no legitimate ERC-20 token with that symbol on Ethereum mainnet — ETH is the native asset, not a token contract — so both are impersonating, yet neither would be flagged by any check for anomalous characters. Four distinct contracts across the trace present the symbol `stETH` in clean ASCII; only one is the Lido contract.

The only reliable identifier of a token is its contract address. *Every token referenced in this investigation is identified by contract; the symbol is recorded as displayed but carries no evidentiary weight.*

8.2 Address poisoning

The impersonating tokens were not sent to arbitrary addresses.

Lookalike addresses generated at scale

OBSERVED Of 124 destination addresses receiving impersonating tokens at the anchor, **31 share both the first six and the last four hexadecimal characters** with an address that received legitimate assets. Twenty-four of the 42 legitimate destinations have at least one such counterpart. FACT

INTERPRETATION Addresses were generated to match the leading and trailing characters of genuine counterparties, then used as destinations for worthless transfers so as to appear in the target's transaction history. Because addresses are conventionally verified by their first and last characters — which is how explorers abbreviate them and how most users check them — the substitution is not visible without full-string comparison. HIGH : thirty-one matches on a twenty-hexadecimal-digit constraint do not arise by chance; each requires deliberate key generation.

BOUNDARY The identity and purpose of whoever deployed these contracts and generated these addresses is **not established**. Deliberate obstruction of forensic analysis, opportunistic phishing directed at the address owner, and automated spam targeting notable addresses all produce this signature. No attribution is made.

The branch selected at Hop 1 has three lookalikes. All four addresses begin `0xa4b2` and end `449e` :

Role	Address
Legitimate — received stETH and mETH	`0xa4b2fd68593b6f34e51cb9edb66e71c1b4ab449e`
Lookalike — received impersonating tokens	`0xa4b24033f4c6d70133e4557c2c32a3fb693e449e`
Lookalike — received impersonating tokens	`0xa4b27b89345db2e5e1a7d7bc537fac354b50449e`
Lookalike — received impersonating tokens	`0xa4b2c2e7a2a0f7f406257d992af5e6bca5d0449e`

8.3 The trap the two techniques form together

A SILENT FAILURE MODE

The impersonating `stETH` (contract `0xee33058e…c562`) shows **554,882 units outbound — the largest single figure in the entire retrieval**, larger than any legitimate movement. Its destination is `0xa4b24033…449e` , a lookalike of the genuine branch.

An analyst who ranked branches by displayed amount without verifying contract addresses would have selected a **fabricated flow**, followed it to a **lookalike address**, and continued from there. Every subsequent hop would be wrong. **Nothing in the output would indicate an error.**

Controls applied. Branch selection is performed on legitimate assets only, verified by contract address. Destination addresses are compared in full, never by leading and trailing characters. Retrieval tooling groups transfers by *(symbol, contract)* rather than by symbol, prints the contract for every group, and flags symbols containing non-Latin characters with their Unicode code points made explicit — so that a reader can see what is being impersonated rather than having to trust that two identical-looking strings differ.

Residual limitation. Character-anomaly detection catches crude impersonation. A contract deployed with a clean ASCII symbol and a false claim to being a known asset passes unflagged, as Hop 3 demonstrated. **Confirming that a contract is the asset it claims to be remains an analyst task**, performed against independent sources. The tooling narrows the problem; it does not solve it.

9. Observed Typologies

Six movement patterns were observed across the trace. Each is examined below against the FATF 2020 Red Flag Indicators, consulted directly at source.

9.1 Method

Two rules govern this section. **The pattern is observed before it is named:** each entry begins from what the blockchain records, then asks whether that behaviour corresponds to a catalogued typology — never the reverse. **Naming a typology is an inference, not a fact:** that funds moved in a certain way is a fact; that the purpose was to launder, obscure, or evade is an interpretation carrying a confidence level and its alternative explanations.

Four of the six patterns have no direct correspondence in the framework, and are recorded as analytical observations of this study rather than forced into a classification. That two correspond directly and four do not is itself evidence that the classification was driven by the evidence rather than by a wish to find red flags everywhere.

The framework's own caveat governs the whole section: the FATF states (§§7–10, pp. 5–6) that a single indicator is not necessarily sufficient to suspect money laundering, that indicators must be read in context, and that suspicion increases where **multiple indicators appear together without a logical economic or business explanation**. Every entry below is therefore accompanied by the legitimate alternative that the observation alone cannot exclude.

9.2 Pattern 1 — Transactional fragmentation

Fragmentation at three levels

OBSERVED Anchor: 400,000 ETH → 40 identical parcels of 10,000, in ~6 minutes. Hop 3: 90,000 ETH → 9 identical parcels of 10,000, in ~48 seconds. Hop 4: 10,062 ETH → 74 transfers of irregular amounts, over ~2h 15m. FACT

FATF **Direct correspondence.** §11, *Size and frequency of transactions*, p. 6: *"Making multiple high-value transactions in short succession, such as within a 24-hour period."* The forty transfers in six minutes and the nine in forty-eight seconds fall squarely within this indicator. HIGH

NOT CLAIMED This is **not structuring** in the sense the FATF reserves for that term. Structuring divides value to remain below a reporting or recording threshold. Parcels of 10,000 ETH are not below any threshold; the mechanism the FATF associates with structuring is absent, and the term is deliberately not used.

SUB-FINDING Fragmentation at the anchor and Hop 3 is uniform and round; at Hop 4 it is irregular. The defensible observation is a **change of mechanism between levels** — not that the Hop 4 pattern is "organic". Irregular amounts are equally consistent with a second automated distribution rule (proportional calculation, residual balances, batching). MODERATE

ALTERNATIVE Dividing funds across many addresses is routine custody practice: hot/cold separation, per-address security, operational batching, internal balance policies. Fragmentation alone is not an incriminating finding.

9.3 Pattern 2 — Conversion of staking derivatives to native ETH

Multi-asset conversion within 50 minutes

OBSERVED stETH, mETH and cmETH converted to native ETH through three separate decentralised exchanges, within approximately 50 minutes of the anchor's outbound activity. FACT

FATF **Direct correspondence.** §12, *Transactions concerning all users*, p. 9: *"Converting a large amount of fiat currency into VAs, or a large amount of one type of VA into other types of VAs, with no logical business explanation."* HIGH

INTERPRETATION The conversion may serve to consolidate into an asset with no protocol issuer: native ETH cannot be frozen or paused by any party, whereas protocol-issued tokens can, depending on design, be subject to issuer-level controls. LOW — a hypothesis about motivation. The FATF indicator concerns the conversion itself, not any purpose behind it, and the blockchain does not record intent.

ALTERNATIVE **Strong.** Conversion to native ETH is a routine prerequisite for almost any subsequent on-chain activity: gas payment, liquidity, and interaction with most services all require it.

9.4 Pattern 3 — Dormancy between stages

Two addresses dormant for the same window

OBSERVED Two distinct addresses received funds on 2025-02-21 and remained inactive approximately nine to ten days before moving them onward — one on 2025-03-02, the other on 2025-03-03. FACT

FATF **No correspondence identified.** The 2020 indicators contain none for which a delay between receipt and onward movement is by itself a red flag. §11 mentions a *"staggered and regular pattern, with no further transactions recorded during a long period afterwards"* — this was considered and found **not** to match: that indicator describes staggered regular transactions followed by silence, while the observation here is the opposite shape, complete dormancy followed by a burst. The text was read and judged inapplicable rather than stretched to fit.

INTERPRETATION The near-simultaneous dormancy of two separate addresses over the same window is the notable feature — more consistent with coordinated handling than with coincidence. LOW : simultaneity is suggestive, and two data points do not establish a pattern.

ALTERNATIVE Waiting for liquidity, market conditions, operational availability, and custody schedules all produce dormancy of this kind.

9.5 Pattern 4 — Reconvergence of separated paths

The branching structure is not a tree

OBSERVED Four addresses receive from both Hop 3 and Hop 4, which sit on different levels of the same branch structure. One received a tied branch that was *not* followed and later received again via the branch that *was*. FACT

FATF **No correspondence identified.** The framework addresses multiple accounts, assets, and service providers, but contains no indicator for funds that separate and later reconverge.

INTERPRETATION Fragmentation followed by reconvergence is consistent with the dispersion having functioned as an intermediate transit stage rather than as final distribution. LOW

ALTERNATIVE **Strong.** Two independent paths using the same downstream service produce identical reconvergence, in which case it is a consequence of that service's architecture rather than of any coordination.

9.6 Pattern 5 — Cross-chain movement

Systematic bridge deposits on adjacent branches

OBSERVED Two of the four convergent addresses at Hop 4 deposit into THORChain, a cross-chain swap protocol; one systematically, in amounts of 208–223 ETH, receiving from three addresses in the same labelled cluster. FACT

FATF **Not a 2020 indicator.** Cross-chain movement does not appear among the enumerated indicators in the 2020 Red Flag Indicators document. Stating "FATF 2020 red flag: chain-hopping" would be incorrect. The technique is discussed in *other* FATF material as one that increases transaction anonymity and impedes tracing.

Formulation used: *the observed behaviour is consistent with chain-hopping, a technique discussed by the FATF in documents subsequent to the 2020 Red Flag Indicators, but it does not constitute one of the specific indicators enumerated in that report.*

SIGNIFICANCE A cross-chain bridge is the primary termination condition of this investigation's stopping rule, because it severs the on-chain link between input and output for open-source analysis.

ALTERNATIVE Bridges and cross-chain swaps have entirely legitimate uses; interoperability is a normal ecosystem feature. Their use is not, by itself, evidence of laundering.

9.7 Pattern 6 — On-chain analytical obstruction

A category of its own

OBSERVED Impersonating token contracts and lookalike destination addresses at every hop, including contracts first appearing only at Hop 5 (Section 8). FACT

FATF **No correspondence, and none should be forced.** No indicator describes address poisoning or token impersonation, and the phenomenon should not be reclassified as laundering to fit the framework. The distinction is categorical: **laundering concerns the movement and transformation of value; impersonation and poisoning concern the manipulation of the informational environment** — inducing error, confusing attribution, degrading analysis. One moves money; the other attacks the analyst.

TREATMENT Recorded under a category of this study's own: **on-chain analytical obstruction**, with the explicit statement that no direct correspondence was identified in the framework consulted.

INTERPRETATION Consistent with deliberate contamination of the transaction record to impede tracing. MODERATE as to the effect — the impersonation and poisoning are established facts, and their effect of impeding analysis is demonstrable. "Deliberate" is the inference: spam and opportunistic phishing produce the same artifacts.

BOUNDARY The authorship of this activity is **not attributable** with the available data.

9.8 Summary

#	Pattern	FATF 2020 correspondence	Treatment
1	Transactional fragmentation	§11, p. 6 — multiple high-value transactions in short succession	**Direct** red flag
2	Derivatives → native ETH	§12, p. 9 — conversion between VAs without logical business explanation	**Direct** red flag
3	Dormancy of 9–10 days	None (§11 considered and found inapplicable)	Analytical observation
4	Reconvergence	None	Structural observation
5	Cross-chain movement	Not in the 2020 indicators; discussed in later FATF material	Consistent with chain-hopping; not a 2020 red flag
6	Impersonation and poisoning	None	Own category: analytical obstruction

The pattern that matters is the combination. The chain observed — large holdings, immediate high-value fragmentation, conversion of derivatives to an unfreezable native asset, multi-level fragmentation with a changing mechanism, dormancy, reconvergence, and eventual approach to a cross-chain bridge — presents several such indicators in sequence. This is recorded as a description of the aggregate pattern, at MODERATE confidence, and **not** as a determination of illicit activity, which the evidence available to this analysis cannot establish. Each legitimate alternative above remains on the record.

10. Evidence Reliability Assessment

Every substantive claim produced by this investigation, classified by type and — where applicable — confidence. The register below was finalized and committed **before** any published analysis was consulted.

10.1 Facts — established on-chain

Direct reads of the ledger, verifiable by any party against any node. No confidence level attaches; these are not inferences.

#	Claim
F1	The anchor made outbound transfers of approximately 400,000 ETH plus staking derivatives on 2025-02-21
F2	90,375.55 stETH, 8,000 mETH and 1 ETH were transferred from the anchor to `0xa4b2fd68…449e`
F3	`0xa4b2fd68…449e` sent stETH and mETH to three contracts and 98,048.79 native ETH to `0xdd90071d…5f92`
F4	The three derivative-receiving destinations are contracts, not accounts
F5	`0xdd90071d…5f92` sent 90,000 ETH in nine transfers of exactly 10,000 ETH within 48 seconds
F6	The earliest of those nine (16:04:23) went to `0xf3025725…72be`
F7	31 addresses receiving impersonating tokens share the first six and last four characters of an address receiving legitimate assets
F8	Four addresses receive from both Hop 3 and Hop 4
F9	`0x327ffe25…ba23` sent 1,651.95 ETH while the traced branch delivered 454.06 ETH to it
F10	Fragmentation occurred at three levels, of 40, 9 and 74 branches respectively

10.2 Inferences — with confidence and basis

#	Claim	Confidence	Basis and reasoning
I1	The three Hop 2 contracts are swap venues, and the derivatives were converted to native ETH there	HIGH	Two independent lines converge: inflow and native outflow reconcile to 0.333%, and all three contracts are verified as DEX infrastructure. Neither alone would be conclusive; together they leave no credible alternative
I2	The fragmentation corresponds to the FATF indicator on multiple high-value transactions	HIGH	The indicator was verified at source and the observed windows fall squarely within it. Applying a framework is nonetheless an interpretive act, not a direct ledger read — hence inference, not fact
I3	The change from round to irregular amounts reflects a change of fragmentation *mechanism*	MODERATE	The contrast is directly observed; that it reflects two distribution rules is the reasonable reading, but a single rule with a residual-balance stage would produce the same
I4	The traced path underwent layering	MODERATE	Consistent with multi-level fragmentation, conversion and delay in sequence; but layering is a function ascribed to the behaviour, and the legitimate-custody alternative is not excluded
I5	The near-simultaneous dormancy reflects coordinated handling	LOW	Simultaneity is suggestive; liquidity, market timing and operational schedules produce the same, and two data points do not establish a pattern
I6	Reconvergence indicates common control of the addresses	LOW	An inference about a *property of the addresses*, **not** an attribution to any named actor. Rated Low because two independent paths using the same downstream service produce identical reconvergence
I7	Conversion to native ETH served to consolidate into an unfreezable asset	LOW	A hypothesis about motivation. Native ETH has no issuer-level freeze, but conversion is also a routine prerequisite for gas and most services
I8	Impersonation and poisoning were deliberate contamination of the record	MODERATE	The artifacts are facts (F7); their effect of impeding analysis is demonstrable. "Deliberate" is the inference — spam and opportunistic phishing produce the same signature

10.3 Attributions — recorded, not asserted

These were encountered and informed navigation. **This analysis endorses none of them.**

#	Claim	Source and status
A1	`0xf3025725…72be` is "Bybit Exploiter 45" and associated with the incident	Block explorer, citing a named third-party investigator. Not an on-chain fact and not a finding of this analysis
A2	`0x21032176…044c` is "Bybit Exploiter 46", funding "Bybit Exploiter 47"	Block explorer; sequential labelling across the cluster
A3	The anchor is "Bybit Exploiter"	Block explorer; the basis for anchor selection, validated on-chain by behaviour rather than accepted on the label

> **WHY THESE ARE NOT FACTS**
>
> A label reading "Bybit Exploiter 45" is an assertion by the explorer, itself relaying a third-party investigator. Nothing on-chain establishes that an address belongs to a named actor. Treating such a label as a fact would be the on-chain equivalent of treating a user account as proof of a physical person's identity — a category error. Where this report refers to these labels, it does so in the form *"[source] associates this address with the incident,"* never *"this address belongs to [actor]."*

10.4 The private attribution boundary

The point at which open-source analysis ceased to be sufficient. This is a stated objective of the investigation (Section 2.1), not a shortfall.

Form	Nature of the boundary
Traceability dissolution Hop 5	A consolidation point received 454 ETH by the traced path and forwarded 1,651 ETH assembled from several inbound paths. Because ETH is fungible, no on-chain evidence distinguishes which outbound transfer carries the traced funds. **A limit of the evidence, not of the tooling** — the information required does not exist on-chain, and no commercial platform or additional effort resolves it.
Cross-chain bridge exposure adjacent to Hop 4	Addresses one step to either side of the traced path deposit into a cross-chain bridge, which severs the on-chain link between input and output: funds enter on Ethereum and leave on another chain with no on-chain record joining the two. **This is where a commercial platform with proprietary cross-chain heuristics would continue and this analysis cannot** — the boundary between open-source and proprietary capability that the investigation set out to locate.

11. Cross-Validation Against Published Analyses

The findings of Sections 7–10 were committed to version control before any published analysis was read. They are **not revised in this section.** Revising them now would destroy the measurement this comparison exists to produce: how far an open-source analyst reaches independently, and where that reach ends relative to parties with proprietary tools.

The purpose is not to check whether the trace was "right". It is to locate, by contrast with published work, the boundary between what open-source analysis established and what required capabilities this investigation did not have.

11.1 Sources consulted

#	Source	Author / date	Grading	Basis for grading
S1	On-chain fund tracing analysis, read via republication	SlowMist Security Team, 2025-02-23	**B2**	Competent specialist firm with detailed analysis. Rated B rather than A because it was read via a republication rather than the primary publication, and because its tracing relies on a proprietary tool whose heuristics cannot be independently verified from public data
S2	*In-Depth Technical Analysis of the Bybit Hack*	NCC Group — Rivas, Santos & Sanz, 2025-03-10	**A2**	Named authors; primary technical detail including contract addresses and code, independently checkable on-chain
S3	FBI public service announcement, as reported	Associated Press — Gambrell, 2025-02-27	**A1**	A1 for the fact of the FBI statement. The attribution it contains is the FBI's, not this analysis's

The B2 grading of S1 is not a criticism of the source. It is the same standard applied to every source in this investigation: what could not be verified at first hand is graded accordingly.

11.2 Convergence

The specialist analysis (S1) traces the same initial path this investigation reconstructed. The correspondence is transfer-by-transfer.

Element	Published (S1), 2025-02-23	This analysis, frozen	Result
Anchor address	`0x47666Fab…86E2`	`0x47666Fab…86E2`	✔ identical
Asset composition	401,347 ETH · 90,375.5479 stETH · 8,000 mETH · 15,000 cmETH	400,001 ETH observed leaving anchor; stETH 90,375.55, mETH 8,000, cmETH 15,000	✔ identical to four decimals on stETH
Anchor fragmentation	400,000 ETH → 40 addresses, 10,000 each	40 × 10,000 ETH (Hop 1)	✔ identical
Derivative destination	8,000 mETH + 90,375.5479 stETH → `0xA4B2Fd68…449e`	Same address (Hop 1 → Hop 2)	✔ identical
Conversion venues	Converted to 98,048 ETH via two named DEX protocols	Converted via three DEX contracts verified by behaviour (Hop 2)	✔ same protocols
Post-conversion destination	→ `0xdd90071d…5f92`	→ `0xdd90071d…5f92` (Hop 2)	✔ identical
Second fragmentation	→ 9 addresses, 10,000 ETH each	9 × 10,000 ETH (Hop 3)	✔ identical
cmETH path	15,000 cmETH → `0x1542368a…4443`	Recorded as an unfollowed branch at Hop 1, same destination	✔ identical

> **THE CENTRAL CORROBORATION**
>
> **This investigation independently reconstructed the published trail through Hop 3, using only a public block explorer, without having read that analysis.** Every address, every amount, and every structural feature — the forty-way split, the derivative consolidation, the DEX conversion, the nine-way split — matches.

Two points of independent corroboration deserve note.

Protocol identification by behaviour. At Hop 2 this analysis verified a conversion contract, found it carrying a protocol tag it did not recognise, and classified it by behaviour as a DEX aggregator. The published analysis names the venue by that protocol's former name. The two arrived at the same protocol — this analysis by verifying behaviour rather than trusting the label, which is the method the investigation set out to apply, and which held even where the label had changed.

Precision of the match. The stETH figure (90,375.5479) agrees to four decimal places between an independent public-explorer trace and a specialist firm's proprietary-tool analysis. A match at that precision is not coincidence; it confirms both were reading the same on-chain events.

11.3 Divergence, and the boundary it locates

The published sources continue past the point where this investigation stopped. **This is the central result of the comparison, and it is not a shortfall.**

Where this analysis stopped: at traceability dissolution (Hop 5) and, on adjacent branches, at a cross-chain bridge. **Where the sources went:** S1 reports a portion of the funds exchanged for Bitcoin through a cross-chain protocol and transferred to a Bitcoin address; S3 reports the FBI stating that the actors converted assets to Bitcoin and other assets dispersed across thousands of addresses on multiple blockchains.

Capability	This analysis	Specialist firm (S1)	FBI (S3)
Ethereum on-chain tracing	✔	✔	✔
Cross-chain (ETH → BTC) tracing	✘ stops at the bridge	✔ via proprietary heuristics	✔
Attribution to a named actor	✘ *by design*	Inferred via cross-incident clustering	✔ via off-chain intelligence and legal process

WHAT THE DIVERGENCE ESTABLISHES

The divergence is not error. Open-source analysis and the proprietary platform agree **completely** on the Ethereum-native path. They part precisely at the cross-chain boundary — where proprietary cross-chain heuristics continue and a public explorer cannot follow. This is the private-attribution boundary the investigation set out to find, measured against the specific class of tool that crosses it.

11.4 What the sources revealed about frozen findings

Two findings recorded at low or moderate confidence are addressed by the sources. **These are not revisions** — the frozen findings stand as written. They are recorded as the outcome of comparison.

The cmETH branch — corroborating inference I7

FROZEN FINDING 15,000 cmETH recorded as an unfollowed branch (Hop 1). Separately, I7 hypothesised at LOW confidence that conversion to native ETH may have served to reach an asset with no issuer-level freeze.

SOURCE S1 reports that the issuing protocol suspended cmETH withdrawals and **recovered the 15,000 cmETH** from the address.

OUTCOME Direct evidence for the mechanism behind I7: an issuer *can* freeze and reclaim a protocol asset, and here one did. The branch this analysis did not follow terminated by issuer intervention. **I7 is corroborated** — though it remains an inference about motivation, not a proven purpose.

Actor attribution — confirming the decision not to attribute

FROZEN FINDING The explorer labels were recorded strictly as third-party attributions (A1–A3). No attribution was made by this analysis.

SOURCES S1 and S3 attribute the incident to a named state-linked actor — S1 via wallet-cluster correlation with addresses from prior incidents and a third-party investigator's evidence (test transactions, associated wallets, timing analysis); S3 via the FBI's own intelligence.

OUTCOME **This vindicates the reliability discipline, not the attribution.** These parties reached attribution through evidence this investigation explicitly did not have: cross-incident clustering databases, off-chain intelligence, subpoena power. That this analysis stopped short of attribution was correct — it lacked the basis to make one, and said so.

11.5 Coverage assessment

Metric	Result
Hops independently reconstructed	5
Corroborated by published analysis	Hops 1–3 correspond transfer-for-transfer; Hops 4–5 continue along the same labelled address cluster
Point of departure	The Ethereum → Bitcoin cross-chain boundary
Reason for departure	Cross-chain tracing requires proprietary heuristics or off-chain legal powers; a public block explorer cannot follow value across a bridge
Proportion of the incident traced	A single path ending in 454 ETH — approximately 0.11% of the funds leaving the anchor

Scope of this validation. The convergence demonstrated covers the single path this investigation followed. It does not validate any claim about the remainder, which was never examined. What is validated is narrow and specific: **that the method applied here — public-explorer tracing, contract verification by behaviour, and fact/inference/attribution discipline — produces, on the segment it can reach, results identical to those of a specialist firm using paid tools.** That is the claim this project set out to test, and the comparison supports it.

12. Limitations

Each limitation is stated with its concrete impact on the conclusions, so that its practical significance — not merely its existence — is clear.

Limitation	Impact
A single path was traced	*Bounds every conclusion in this report.* The followed path carries approximately 0.11% of the value leaving the anchor. The remaining 99.89% was not analysed. No statement about "the funds" applies beyond the documented path. This is the stopping rule operating as designed, but it is the report's most consequential boundary.
The cross-chain boundary was not crossed	*Terminates the trail.* Where funds enter a cross-chain bridge, no on-chain record joins input to output. This is not a gap in effort but the point at which the required information ceases to exist on the Ethereum ledger. Section 11 confirms that crossing it requires proprietary heuristics or legal process.
No attribution capability	*Removes an entire class of conclusion.* Cross-incident clustering databases, off-chain intelligence, and subpoena power were unavailable. Consequently no actor is named. Section 11.4 confirms the restraint was correct rather than merely cautious.
Token legitimacy verification is manual	*Affects the reliability of signal separation.* Character-anomaly detection catches crude impersonation; a clean-ASCII forgery passes unflagged, as occurred at Hop 3. Two such contracts were excluded on analyst judgment, not automated warning. A less careful reading would have included them.
Cross-asset valuation assumed 1:1	*Bounds the Hop 1 branch selection.* The derivatives were treated as ETH-equivalent without consulting market data for the relevant time. The selected branch exceeded alternatives by a factor of ~6.5, so the outcome does not turn on the precise rate — but the assumption would not be sufficient where two branches were close in value.
Explorer labels are perishable third-party claims	*Affects reproducibility of verification steps, not of findings.* Contract and address classifications relied partly on labels that may be revised or removed. Captures were taken with observation dates. Transaction data itself is immutable and reproducible indefinitely.
Inbound analysis was out of scope	*Leaves two discrepancies unexplained.* Outflow exceeded traced inflow by 6.57 ETH at Hop 3 and 62.36 ETH at Hop 4, establishing that both addresses received from untraced sources. The magnitudes are small relative to throughput, but the sources are unknown and are recorded rather than assumed away.
Egress control was applied selectively	*Affects the operational-security posture, not the findings.* The free-tier VPN's automatic server assignment produced latency that made continuous use impractical; the control was therefore active for all explorer queries and API calls and disabled during local documentation work. The distinction is recorded so the posture is not overstated.
Single analyst, no peer review	*Affects assurance, not any specific finding.* No independent examiner checked branch selections, signal separation, or classifications. The pre-registered rules, complete branch records, and published evidence data are intended to make every decision auditable by a third party — but that audit has not been performed.
Retrieval is a point-in-time snapshot	*Minimal.* On-chain data is immutable, so transaction records reproduce indefinitely. Balances and labels observed on the retrieval dates will differ later; each is recorded with its observation timestamp.

13. Legal and Ethical Context

Open-source intelligence is bounded less by technical capability than by legal and ethical constraint. The methodology applied here was chosen to remain well inside those boundaries, and the reasoning is recorded so it can be assessed rather than assumed.

13.1 Passive collection and the boundary of lawful access

No port scans, service enumeration, vulnerability probes, or connection attempts were directed at any target infrastructure. No wallet was connected, no contract called, no transaction broadcast. Every query was issued to a public block explorer or its public API.

This distinction is legally material. Consultation of published records does not constitute unauthorized access, whereas active probing of systems without authorization may, under Brazil's Lei 12.737/2012 (Art. 154-A, *invasão de dispositivo informático*) and comparable statutes elsewhere. The method was selected to keep the investigation on the correct side of that line by design, not by accident.

Blockchain ledgers are, further, public by architecture. Reading one is not analogous to accessing a private system; the data is published by the protocol to every participant. This is the most legally secure form of investigation available in this domain.

13.2 Personal data

Under the LGPD and comparable regimes, personal data is information relating to an identified or identifiable natural person. A blockchain address, standing alone, does not identify a natural person, and **no attempt at such linkage was made** — identification of natural persons is explicitly out of scope (Section 2.3).

The principle that does apply is **purpose limitation**: data was collected only to the extent required to trace the fund path, and no enrichment, cross-referencing, or attribution to individuals was attempted. Public availability would not, by itself, authorize unrestricted processing.

13.3 Target selection and disclosure restraint

The subject is an incident already extensively documented in public reporting, involving a major exchange and addresses already publicly labelled. No private individual, small entity, or party with a reasonable expectation of obscurity was examined. The same techniques applied to a private individual would raise materially different ethical questions even where the data is equally public.

A disclosure restraint was fixed in advance: any address, pattern, or relationship surfaced that was **not already present in public reporting** would be withheld from publication and reviewed privately before any disclosure decision. No such material arose — every element documented here appears in, or is derivable from, information already public.

13.4 Evidentiary posture

Should open-source analysis of this kind be used to support a formal proceeding, its weight would depend on **provenance rather than content**. Four properties of this investigation bear on that:

- **Reproducibility.** Every finding ties to a transaction hash on a public ledger. Any party can re-verify any transaction recorded here against any node, indefinitely. This is a stronger reproducibility guarantee than most forensic domains afford.
- **Documented provenance.** The collection log records the source, query, and timestamp of every external consultation, including the weak source used to locate the anchor and the point at which external reading stopped.
- **Pre-registration and sequence integrity.** The stopping rule was committed before data was examined, and the findings before any published analysis was read. Both are evidenced by the version-control history, which makes the sequence auditable rather than asserted.
- **Disclosed limitations.** Every deviation and every unexplained discrepancy is stated (Section 12). Undisclosed limitations, if later discovered, are far more damaging to credibility than disclosed ones.

Perishability, and why it matters here. Open-source data of this kind divides sharply. Transaction records are immutable and reproduce indefinitely. Entity labels, explorer annotations, and platform attributions are revised without notice. Every label relied upon in this report is therefore recorded with its platform and observation date, and captured

— because contemporaneous provenance documentation, not the finding itself, is what makes such material defensible later.

Under Brazilian procedure, technical examination of this kind would inform a *laudo pericial* prepared by a court-appointed or party-retained expert (*perito*), governed by the Código de Processo Penal and Código de Processo Civil rather than by this report's academic format. The discipline demonstrated here — documented provenance, reproducible steps, and explicit separation of fact from inference — is the same discipline that would underpin such an instrument. **This report is not one**, and does not constitute a legal instrument, an accusation, or an intelligence product in any jurisdiction.

14. Conclusions

1. **A fund path was independently reconstructed across five hops.** From the anchor through derivative consolidation, conversion to native ETH via three decentralised exchanges, two levels of fragmentation, and a final consolidation point. Every hop is documented with transaction hashes; every branch not followed is recorded with its address and value.
2. **The reconstruction was corroborated by a specialist firm's published analysis, which was read only after the findings were frozen.** The two agree transfer-for-transfer through the first three hops, with the stETH figure matching to four decimal places. On the Ethereum-native segment, a free block explorer and a proprietary intelligence platform produced identical results.
3. **The boundary of open-source analysis was located precisely.** The two approaches diverge at exactly one point: the cross-chain bridge. Crossing it requires proprietary cross-chain heuristics or legal process. The limit encountered is not one of skill, effort, or diligence — it is the point at which the required information ceases to exist on the public Ethereum ledger. *This was the investigation's stated question, and this is its answer.*
4. **Deliberate contamination of the transaction record was identified and characterised.** Nine impersonating token contracts using non-Latin homoglyphs, clean-ASCII forgeries that no automated check catches, and 31 lookalike addresses matching genuine counterparties in their first six and last four characters. Together these form a trap capable of diverting a trace onto a fabricated path with no visible error — and one that the largest figure in the entire dataset was positioned to spring.
5. **Two FATF red-flag indicators were identified and verified at source; four further patterns had no correspondence and were not forced into one.** Fragmentation (§11, p. 6) and multi-asset conversion (§12, p. 9) correspond directly. Dormancy, reconvergence, cross-chain movement, and analytical obstruction do not, and are recorded as observations of this study.
6. **No attribution is asserted.** Published sources attribute the incident to a named actor on evidence this analysis did not have. The comparison in Section 11 confirms that stopping short of attribution was correct rather than merely cautious.
7. **The trace covers approximately 0.11% of the funds leaving the anchor.** This bounds every conclusion above. Nothing in this report describes the remaining 99.89%, which was not examined.

> **WHAT THIS INVESTIGATION DEMONSTRATES**
>
> That a rigorous open-source method — pre-registered decision rules, provenance-documented sources, contract verification by behaviour rather than by label, and systematic separation of fact from inference from attribution — reproduces professional results on the segment where public data suffices, and **identifies with precision the point at which it no longer does**. Knowing where a method stops working is not a lesser result than following it further. It is the result that tells a reader what any conclusion is worth.

15. Analyst Declaration

I, Paulo Vaz, declare that:

1. This report accurately reflects the findings of the investigation described herein, conducted between 24 and 28 August 2026 on publicly available blockchain data.
2. All data examined is public by design. No query was directed at any private system, no wallet was connected, no contract was called, and no transaction was broadcast at any point.
3. The decision rules governing branch selection and termination were committed to version control before any transaction data was examined, and the findings before any published third-party analysis was consulted. Both sequences are evidenced by the project's version-control history.
4. Every claim is classified as fact, inference, or third-party attribution, and every inference carries an explicit confidence level. Where a pattern had no correspondence in the reference framework, that absence is stated rather than a correspondence constructed.
5. **No attribution to any individual, organization, or state is asserted by this analysis.** Where published sources make such attributions, they are recorded as third-party claims and are not adopted as findings.
6. All limitations, deviations, and unexplained discrepancies are disclosed in Section 12.
7. This report was prepared for academic and portfolio purposes and does not constitute a legal instrument, an accusation, or an intelligence product in any jurisdiction.

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Paulo Vaz — Blockchain Investigation Analyst
Date: _______ / _______ / 2026

Case BIC-2026-001. Investigation conducted 24–28 August 2026; report issued 4 September 2026. Evidence data, retrieval tooling, phase documentation, and the complete version-control history are published in the project repository. Every transaction referenced is verifiable by any party against any Ethereum node, indefinitely.